

Diego Fernando Lojan Tenesaca

Data & AI Engineer | Full Stack Developer

Available for remote work

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PROFESSIONAL SUMMARY

Computer Science Engineer with four years developing software, the last two professionally, and two focused on machine learning, data science, data analysis and natural language processing (NLP). I was part of the team behind the AI microservice of a live product, with multi-tenant architecture, 18 functional domains, a 15-tool agent and semantic search over 139,000 vectorized conversations on pgvector. My work has been taking models to production and keeping them there. I build ETL pipelines, work with RAG and hybrid retrieval, orchestrate LLMs and vision-language models with LangGraph and run accelerated inference on GPU and on-device with ONNX Runtime, including face recognition and video analysis. I also develop the web and mobile applications that consume all of it. I have a paper accepted at Springer Nature (CIT 2026) on stress detection in Spanish.

PUBLICATIONS

- **Lojan, D. F.; Chamba-Eras, L.** – *Bayesian Hyperparameter Optimization of Support Vector Machines Using TPE for Stress Detection in Spanish Social Media During the 2024 Ecuadorian Energy Crisis*. In *Emerging Research in Intelligent Systems*, proceedings of the Congress on Innovation and Technology (CIT 2026), *Lecture Notes in Networks and Systems* series, Springer Nature. (Accepted, in production)

PROFESSIONAL EXPERIENCE

Indaga Lab | *Founder & Lead Developer*
Quito, Ecuador / Remote

2025 – Present
Own software and AI product studio

- **Wauto Indaga:** AI-powered CRM and messaging automation for WhatsApp, Telegram and Meta, live in production with five commercial tiers and agent replies under 2 seconds. Hybrid retrieval (full-text + pgvector), model cascade with failover, versioned public API and MCP server.
- **Kumbre:** converts construction bids from PDF (Ecuadorian public procurement format) into editable Excel through intelligent parsing and fuzzy matching, with a preloaded catalog and official price indexes.
- **AgentOS:** Android app that turns Google-free phones into a 24/7 AI agent operated over Telegram. Released as open source.
- Technical leadership of the team, product definition and pricing model.

Freelance Developer | *Software & AI Solutions*
Remote

2024 – Present

- Conversational bots (Telegram, WhatsApp) with LLM orchestration, cross-provider failover, versioned prompts and strict output validation. Lead qualification, CRM and customer support flows.
- CRM for an energy advisory firm, plus shared-account management platforms with automatic OTP extraction, email webhooks and an admin panel.
- Anomaly detection over multi-country time series, data feasibility studies with web scraping, and data automation with n8n.
- Computational support for theses and academic research: end-to-end numerical simulation, MATLAB portability and reproducible NLP experiments.

Dataglov S.A.S. | *Junior Software Engineer (Data & AI)*
Loja, Ecuador

Feb 2026 – Aug 2026
AI microservice in production

- AI microservice built on DDD with schema-isolated multi-tenancy in PostgreSQL, organized into **18 functional domains** behind a single REST API.
- Semantic search over **139,000 vectorized conversations** using multilingual BGE-M3 embeddings (1,024 dimensions) and an HNSW index in pgvector.
- ETL pipelines with async jobs and background workers. Re-embedded **62,000 conversations in under an hour** on an NVIDIA A6000 GPU with Text Embeddings Inference.
- In-house benchmark of vector engines across the full 139,000 real vectors (latency, recall and memory) to pick the production engine.
- Conversational agent with **15 tool-calling tools**, loop control, normalized caching and PII masking, orchestrated with LangChain and LangGraph.
- Self-hosted Vision-Language Model pipelines on GPU (Qwen-VL, InternVL) with batch processing, checkpoints and retries. Zero-shot classification, OCR, local audio transcription and offline translation services.

- Reversible migrations with checksums, model observability (token and latency tracing), per-tenant rate limiting and containerized deployment with CI/CD.

Serviestudios Cía Ltda. | *Full Stack Developer*
Loja, Ecuador

Oct 2025 – Feb 2026
Multi-tenant SaaS (web + mobile)

- Next.js web back office and cross-platform React Native mobile app, with dynamic theming and internationalization.
- Domain-partitioned Python REST API with JWT + OAuth authentication and a granular role and permission model.
- PostgreSQL query optimization, real-time messaging over WebSockets and Redis caching.
- Third-party integrations (WhatsApp Business API, Google Calendar, payment gateways, OTP codes), a visual conversational flow builder and OCR document processing.

FEATURED PROJECTS

- **Sudial AI** – Production AI microservice with a domain-partitioned API, an embedding base over real conversations, a tool-orchestrating agent and observability. FastAPI, LangGraph, pgvector.
- **Wauto Indaga** – AI-powered CRM and messaging automation for WhatsApp, Telegram and Meta: multi-agent shared inbox, Kanban pipelines, visual automation builder, hybrid knowledge base and MCP server. Next.js 16, React 19, Supabase.
- **CConnect** (open source, as contributor) – Mobile, desktop and web client for driving a remote coding agent over the Agent SDK, with interactive permissions and secure remote access through Tailscale. A single Compose Multiplatform codebase targets Android, desktop and WebAssembly. Python (FastAPI), Kotlin.
- **Kumbre** – Vertical tool for construction firms: public-procurement PDF bids into editable Excel through intelligent parsing and fuzzy matching, with a preloaded catalog, accounts and plans. Next.js, FastAPI, PostgreSQL.
- **LATAM Energy Anomalies** – Energy crisis detection across 8 countries through multi-technique consensus (Isolation Forest, STL, CUSUM), cross-validation and an interactive dashboard, with an F1 of 0.75. Scikit-learn, Optuna, SHAP, Streamlit.
- **Energy Advisory CRM** (client work, private repository) – Client and case management for an energy sector consultancy: dynamic forms, pagination, state control and reusable components.

Full project catalog, including AgentOS, Reps, OfficeGo and ADBox: [view portfolio](#).

EDUCATION

Computer Science Engineer – Universidad Nacional de Loja	Loja, Ecuador
International Studies in Systems & Computer Engineering (1 term) – UTP	Lima, Peru
Diploma in Project-Based Learning (120 h) – Politécnico de Colombia	Online, 2026

TECHNICAL SKILLS

AI / ML: LLMs (LangChain, LangGraph) | RAG and hybrid retrieval | Tool-calling agents | MCP | Embeddings and Vector Search | NLP | Zero-shot classification | Anomaly Detection | OCR | Speech-to-Text | Vision-Language Models | ONNX Runtime
Data: SQL | ETL | EDA and Data Wrangling | Time Series | Web Scraping | Statistical Analysis | Pandas | NumPy | SciPy | Streamlit | Plotly | MATLAB
Backend & Databases: Python (FastAPI, Pyramid, Django) | Node.js | Java (Spring Boot) | REST APIs | Microservices | WebSockets | OAuth 2.0 / JWT | PostgreSQL (+pgvector) | Redis | Supabase | SQLAlchemy
Frontend & Mobile: React | Next.js | TypeScript | Tailwind CSS | Astro | Nuxt / Vue | PWA | Kotlin | Jetpack Compose | Compose Multiplatform | React Native / Expo | Tauri (Rust)
DevOps & Tooling: Docker | Docker Compose | CI/CD (GitHub Actions, GitLab) | Linux | Git | Claude Code | Agent SDK | MCP servers | n8n

CERTIFICATIONS

Python for Data Science – LinkedIn Learning (2026): [Advanced Data Scientist](#), [Big Data Essentials](#), [PySpark](#), [NLTK](#).
Generative AI & Agents – LinkedIn Learning (2026): [AI Agent Foundations \(Architectures, Frameworks and Applications\)](#), [Generative AI Foundations](#), [What Is Generative AI](#), [Programming Foundations: AI](#).
Data Analysis – Microsoft & LinkedIn (2026): [Career Essentials in Data Analysis](#) (5-course path), [Microsoft Fabric](#).

LANGUAGES

Spanish: Native | English: Intermediate (B1), fluent technical reading and written communication ([view certificate](#))